PAVAN PANDYA

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EDUCATION

Indiana University Bloomington **August 2023 – May 2025**
Master of Science in Computer Science | ***GPA: 3.48 / 4.00*** *Bloomington, Indiana*
Relevant Courses: Applied Algorithms, Applied Machine Learning, Software Engineering, Engineering Cloud Computing

LDRP Institute of Technology and Research **July 2019 – June 2023**
Bachelor of Engineering in Computer Engineering | ***GPA: 4.00 / 4.00*** *Gandhinagar, Gujarat*

TECHNICAL SKILLS

Programming Skills – Python, R, SQL (MongoDB, PostgreSQL, MySQL)
Frameworks – Pytorch, Tensorflow, Keras, OpenCV, Langchain, PySpark, Hadoop, Flask, Django, Fast API
Libraries – Scikit-Learn, Pandas, NumPy, NLTK, SciPy, Matplotlib, Seaborn, Selenium, Huggingface, Regex
Tools – Git, Github, AWS, MLflow, ETL Tools, MLOps, Jira, Linux, Docker, Tableau, Power BI, REST API, Postman, MS Excel

PROFESSIONAL EXPERIENCE

Hate Speech and Offensive Content Identification | *Student Organizer* June 2021 -- Present

- Engineered a REST API backend for a submission platform, facilitating seamless participant submissions & dashboard display with filters based on accuracy metric & task categories, resulting in a 40% increase in annotation throughput.
- Led the collection of 1200 Gujarati tweets for hate speech detection using selenium, enhancing model accuracy by 15% through manual annotation (~ 200 tweets) and fine-tuning using few-shot learning techniques.

Dhirubhai Ambani Institute of Technology | *Research Intern* Nov 2022 – June 2023

- Spearheaded the collection of over 1.5 million tweets using Twitter API and Selenium, and optimized storage in MongoDB with SQL driven schema, showcasing advanced proficiency in data acquisition and database management.
- Utilized advanced text preprocessing techniques and harnessed BERT embeddings to extract inflation-related tweets (~ 157K tweets) from a vast dataset, employing topic modeling libraries such as **BERTopic** and **LDA**.

Brainly Beam Technologies Pvt Ltd. | *Data Science Intern* June 2022 – July 2022

- Developed a recommender system utilizing Linguistics and a Contextual-Based approach for sentiment identification using SVM or Bayes classifier, achieving an accuracy rate of approximately 68% with ML Algorithms.
- Employed NLP methodologies to analyze and predict sentiments associated with generated content, such as reviews and comments, leveraging RNN and LSTM models, to extract insights and enhance content understanding.

PROJECTS

DocQnA – Intelligent PDF Querying LLM System | *Apache Cassandra, Astra DB, LangChain* May 2024 – Present

- Pioneering the development of an advanced PDF Document-based Question Answering System, integrating Retrieval Augmented Generation (RAG) for enhanced accuracy and relevance.
- Streamlined the utilization of Apache Cassandra for efficient data management and integration with Streamlit for a user-friendly interface, revolutionizing information retrieval processes.

Kidney Disease Classification | *Deep Learning, MLflow, DVC, AWS, Docker, GitHub Actions* Nov 2023 – Jan 2024

- Achieved a 15% improvement in kidney disease classification accuracy by developing and fine-tuning ResNet50V2 model. Utilized MLflow for experiment tracking, enabling precise identification of model enhancements.
- Accelerated deployment process time by 30%. Created an IAM user, built a Docker image, pushed it to AWS ECR, launched an EC2 instance, & deployed the image using GitHub Actions, resulting in an efficient AWS CI/CD pipeline.

Uber Data Analysis Pipeline using GCP | *Data Extraction, Data Transformation, Mage AI, Looker* Sept 2023 – Nov 2023

- Orchestrated the development of a robust GCP data pipeline using Python, BigQuery, Looker, and MageAI to analyze NYC taxi trip data, resulting in a 20% improvement in data processing efficiency.
- Enhanced decision-making capabilities by uncovering key patterns and trends in the BigQuery data and translating findings into insightful visual dashboards with Looker. This initiative led to a 15% increase in operational efficiency.

ACHIEVEMENTS

- IBM Hack Challenge 2021-Winner | AIR-16 IIT MADRAS Hackathon | Top-10 Best Student Project – Postman API Fest